CORE PURE / DIFFERENTIAL EQUATIONS

Integrating Factor Method

Worked solutions 15 questions · 110 marks

Questions

Question 1	*7 marks*	2
Question 2	*7 marks*	4
Question 3	*7 marks*	6
Question 4	*6 marks*	8
Question 5	*5 marks*	9
Question 6	*8 marks*	11
Question 7	*6 marks*	13
Question 8	*6 marks*	15
Question 9	*9 marks*	17
Question 10	*9 marks*	20
Question 11	*5 marks*	22
Question 12	*11 marks*	24
Question 13	*8 marks*	27
Question 14	*8 marks*	29
Question 15	*8 marks*	32

1. **(a)** Find the general solution of the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2x}{x^2+1}y = 6x$$

giving your answer in the form $y = f(x)$ **[4]**

(b) Given that $y = \frac{7}{2}$ when $x = 0$, find the smallest positive value of x for which $y = 4$ **[3]**

Solution

(a) This is a linear differential equation of the form $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$ with $P(x) = \frac{2x}{x^2+1}$.

The integrating factor is

$$\begin{aligned}\mu(x) &= e^{\int P(x)\,\mathrm{d}x} \\ &= e^{\int \frac{2x}{x^2+1}\,\mathrm{d}x} \\ &= e^{\ln(x^2+1)} \\ &= x^2+1\end{aligned}$$

Multiplying the differential equation by x^2+1 gives

$$(x^2+1)\frac{\mathrm{d}y}{\mathrm{d}x} + 2xy = 6x(x^2+1)$$

The left-hand side is the derivative of $(x^2+1)y$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\big((x^2+1)y\big) = 6x(x^2+1)$$

Integrating both sides:

$$\begin{aligned}(x^2+1)y &= \int 6x(x^2+1)\,\mathrm{d}x \\ &= \int (6x^3+6x)\,\mathrm{d}x \\ &= \frac{3}{2}x^4 + 3x^2 + c\end{aligned}$$

Hence the general solution is

$$y = \frac{\frac{3}{2}x^4 + 3x^2 + c}{x^2+1}$$

(b) Using $y = \frac{7}{2}$ when $x = 0$:

$$\begin{aligned}\frac{7}{2} &= \frac{\frac{3}{2}(0)^4 + 3(0)^2 + c}{(0)^2+1} \\ \frac{7}{2} &= c\end{aligned}$$

So the particular solution is

$$y = \frac{\frac{3}{2}x^4 + 3x^2 + \frac{7}{2}}{x^2+1}$$

Now set $y = 4$:

$$4 = \frac{\frac{3}{2}x^4 + 3x^2 + \frac{7}{2}}{x^2 + 1}$$

Multiply through by $x^2 + 1$:

$$4(x^2 + 1) = \frac{3}{2}x^4 + 3x^2 + \frac{7}{2}$$

Multiply by 2:

$$8x^2 + 8 = 3x^4 + 6x^2 + 7$$

Rearranging:

$$0 = 3x^4 - 2x^2 - 1$$

Let $u = x^2$. Then

$$3u^2 - 2u - 1 = 0$$

$$(3u + 1)(u - 1) = 0$$

So

$$u = 1 \quad \text{or} \quad u = -\frac{1}{3}$$

Since $u = x^2 \geq 0$, we take $x^2 = 1$, giving

$$x = \pm 1$$

The smallest positive value is

$$x = 1$$

2. **(a)** Solve the differential equation

$$(1+x)\frac{\mathrm{d}y}{\mathrm{d}x} - 2y = (1+x)^4, \qquad x > -1$$

giving your answer in the form $y = f(x)$ **[4]**

(b) The solution curve passes through the point $(0, -1)$. Find the exact positive value of x for which $y = 0$ **[3]**

Solution

(a) Rewrite the differential equation in linear form by dividing through by $1+x$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{2}{1+x}y = (1+x)^3$$

This is a linear differential equation of the form $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$ with $P(x) = -\frac{2}{1+x}$.

The integrating factor is

$$\begin{aligned}\mu(x) &= e^{\int P(x)\,\mathrm{d}x} \\ &= e^{\int -\frac{2}{1+x}\,\mathrm{d}x} \\ &= e^{-2\ln(1+x)}\end{aligned}$$

Since $x > -1$, we have $1 + x > 0$, so

$$\mu(x) = (1+x)^{-2}$$

Multiplying the differential equation by $(1+x)^{-2}$ gives

$$(1+x)^{-2}\frac{\mathrm{d}y}{\mathrm{d}x} - 2(1+x)^{-3}y = (1+x)$$

The left-hand side is the derivative of $y(1+x)^{-2}$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(y(1+x)^{-2}\right) = 1+x$$

Integrating both sides:

$$\begin{aligned}y(1+x)^{-2} &= \int (1+x)\,\mathrm{d}x \\ &= x + \frac{x^2}{2} + c\end{aligned}$$

Hence the general solution is

$$y = (1+x)^2\left(\frac{x^2}{2} + x + c\right)$$

(b) The solution curve passes through $(0, -1)$, so substituting into the general solution from part (a):

$$-1 = (1+0)^2\left(\frac{0^2}{2} + 0 + c\right)$$

$$-1 = c$$

Therefore

$$y = (1+x)^2\left(\frac{x^2}{2} + x - 1\right)$$

To find when $y = 0$, set this equal to zero:

$$(1+x)^2\left(\frac{x^2}{2} + x - 1\right) = 0$$

Since $x > -1$, we have $1 + x > 0$, so $(1+x)^2 \neq 0$. Therefore

$$\frac{x^2}{2} + x - 1 = 0$$

Multiply by 2:

$$x^2 + 2x - 2 = 0$$

Solve using the quadratic formula:

$$\begin{aligned} x &= \frac{-2 \pm \sqrt{2^2 - 4(1)(-2)}}{2} \\ &= \frac{-2 \pm \sqrt{4+8}}{2} \\ &= \frac{-2 \pm \sqrt{12}}{2} \\ &= \frac{-2 \pm 2\sqrt{3}}{2} \\ &= -1 \pm \sqrt{3} \end{aligned}$$

The positive value is

$$x = \sqrt{3} - 1$$

3. **(a)** For $x > -1$, determine the general solution of the differential equation

$$(1+x)\frac{\mathrm{d}y}{\mathrm{d}x} + y = \ln(1+x)$$

giving your answer in the form $y = f(x)$ **[5]**

(b) Given that $y = -1$ when $x = 0$, determine the positive value of x for which $y = 0$ **[2]**

Solution

(a) For $x > -1$, divide the differential equation by $1 + x$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{1+x}y = \frac{\ln(1+x)}{1+x}$$

This is a linear differential equation of the form $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$ with $P(x) = \frac{1}{1+x}$.
The integrating factor is

$$\begin{aligned}\mu(x) &= e^{\int P(x)\,\mathrm{d}x} \\ &= e^{\int \frac{1}{1+x}\,\mathrm{d}x} \\ &= e^{\ln(1+x)}\end{aligned}$$

Since $x > -1$, we have $1 + x > 0$, so

$$\mu(x) = 1 + x$$

Multiplying the differential equation by $1 + x$ gives

$$(1+x)\frac{\mathrm{d}y}{\mathrm{d}x} + y = \ln(1+x)$$

The left-hand side is the derivative of $(1 + x)y$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\big((1+x)y\big) = \ln(1+x)$$

Integrating both sides with respect to x:

$$(1+x)y = \int \ln(1+x)\,\mathrm{d}x + c$$

Integrate by parts with

$$u = \ln(1+x), \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$$

so that

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x}, \qquad v = x$$

Using integration by parts:

$$\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$

So

$$\int \ln(1+x)\,\mathrm{d}x = x\ln(1+x) - \int \frac{x}{1+x}\,\mathrm{d}x$$

Writing $\frac{x}{1+x} = 1 - \frac{1}{1+x}$ gives

$$\begin{aligned}\int \ln(1+x)\,\mathrm{d}x &= x\ln(1+x) - \int\left(1 - \frac{1}{1+x}\right)\mathrm{d}x \\ &= x\ln(1+x) - x + \ln(1+x) \\ &= (1+x)\ln(1+x) - x\end{aligned}$$

So

$$(1+x)y = (1+x)\ln(1+x) - x + c$$

Dividing by $1+x$, which is positive since $x > -1$, gives the general solution

$$y = \ln(1+x) + \frac{c-x}{1+x}$$

(b) Use the condition $y = -1$ when $x = 0$.

Substituting into the general solution:

$$-1 = \ln(1+0) + \frac{c-0}{1+0}$$

$$-1 = 0 + c$$

So

$$c = -1$$

Hence

$$\begin{aligned}y &= \ln(1+x) + \frac{-1-x}{1+x} \\ &= \ln(1+x) - 1\end{aligned}$$

Now set $y = 0$:

$$0 = \ln(1+x) - 1$$

$$\ln(1+x) = 1$$

Exponentiating both sides:

$$1 + x = e$$

$$x = e - 1$$

So the required value is

$$x = e - 1 \approx 1.72$$

4. **(a)** Show that

$$\frac{\mathrm{d}}{\mathrm{d}x}\ln(\sec x+\tan x)=\sec x \qquad \textbf{[2]}$$

(b) Hence find the general solution of the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x}-2y\tan x=\sec^3 x$$

where $-\frac{\pi}{2}<x<\frac{\pi}{2}$. **[4]**

Solution

(a)

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\ln(\sec x+\tan x)&=\frac{\sec x\tan x+\sec^2 x}{\sec x+\tan x}\\&=\frac{\sec x(\tan x+\sec x)}{\sec x+\tan x}\\&=\sec x\end{aligned}$$

(b) This is a linear differential equation of the form $\frac{\mathrm{d}y}{\mathrm{d}x}+P(x)y=Q(x)$ with $P(x)=-2\tan x$.

Using the standard result

$$\int\tan x\,\mathrm{d}x=\ln|\sec x|+c$$

the integrating factor is

$$\begin{aligned}\mu(x)&=e^{\int P(x)\,\mathrm{d}x}\\&=e^{\int -2\tan x\,\mathrm{d}x}\\&=e^{-2\ln(\sec x)}\end{aligned}$$

Since $-\frac{\pi}{2}<x<\frac{\pi}{2}$, we have $\sec x>0$, so

$$\mu(x)=\cos^2 x$$

Multiplying through by $\cos^2 x$:

$$\cos^2 x\frac{\mathrm{d}y}{\mathrm{d}x}-2y\cos x\sin x=\sec x$$

The left-hand side is the derivative of $y\cos^2 x$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}(y\cos^2 x)=\sec x$$

Using part (a):

$$y\cos^2 x=\ln(\sec x+\tan x)+c$$

Hence

$$y=\sec^2 x\big(\ln(\sec x+\tan x)+c\big)$$

5. Find the general solution of the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} - y\tan x = \frac{1}{\sqrt{1+\sin x}}$$

where $-\frac{\pi}{2} < x < \frac{\pi}{2}$ **[5]**

Solution

This is a linear differential equation of the form $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$ with $P(x) = -\tan x$.

Using the standard result

$$\int \tan x \,\mathrm{d}x = \ln|\sec x| + c$$

the integrating factor is

$$\begin{aligned}\mu(x) &= e^{\int P(x)\,\mathrm{d}x} \\ &= e^{\int -\tan x\,\mathrm{d}x} \\ &= e^{-\ln(\sec x)}\end{aligned}$$

Since $-\frac{\pi}{2} < x < \frac{\pi}{2}$, we have $\sec x > 0$, so

$$\mu(x) = \cos x$$

Multiply the differential equation throughout by $\cos x$:

$$\cos x\frac{\mathrm{d}y}{\mathrm{d}x} - y\cos x\tan x = \frac{\cos x}{\sqrt{1+\sin x}}$$

$$\cos x\frac{\mathrm{d}y}{\mathrm{d}x} - y\sin x = \frac{\cos x}{\sqrt{1+\sin x}}$$

The left-hand side is the derivative of $y\cos x$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}(y\cos x) = \frac{\cos x}{\sqrt{1+\sin x}}$$

Integrate both sides with respect to x:

$$y\cos x = \int \frac{\cos x}{\sqrt{1+\sin x}}\,\mathrm{d}x + c$$

Let

$$u = 1+\sin x \quad\Longrightarrow\quad \frac{\mathrm{d}u}{\mathrm{d}x} = \cos x \quad\Longrightarrow\quad \mathrm{d}u = \cos x\,\mathrm{d}x$$

Then

$$\begin{aligned}\int \frac{\cos x}{\sqrt{1+\sin x}}\,\mathrm{d}x &= \int u^{-1/2}\,\mathrm{d}u \\ &= 2u^{1/2} \\ &= 2\sqrt{1+\sin x}\end{aligned}$$

Hence

$$y\cos x = 2\sqrt{1+\sin x} + c$$

Therefore the general solution is

$$y = \frac{2\sqrt{1+\sin x}+c}{\cos x}$$

An equivalent form is

$$y = \frac{2}{\sqrt{1-\sin x}} + c\sec x$$

6. **(a)** Show that an appropriate integrating factor for

$$x^2 \frac{\mathrm{d}y}{\mathrm{d}x} + y = 1, \qquad x > 0$$

is $e^{-1/x}$ **[3]**

(b) Hence find the solution of the differential equation

$$x^2 \frac{\mathrm{d}y}{\mathrm{d}x} + y = 1, \qquad x > 0$$

for which $y = 3$ when $x = 1$ **[5]**

Solution

(a) First write the differential equation in the standard linear form by dividing through by x^2:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{x^2} y = \frac{1}{x^2}$$

This is a linear differential equation of the form $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$ with $P(x) = \frac{1}{x^2} = x^{-2}$.
The integrating factor is

$$\begin{aligned} \mu(x) &= e^{\int P(x)\,\mathrm{d}x} \\ &= e^{\int x^{-2}\,\mathrm{d}x} \\ &= e^{-x^{-1}} \\ &= e^{-1/x} \end{aligned}$$

Hence an appropriate integrating factor is $e^{-1/x}$.

(b) Using the integrating factor from part (a), multiply

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{x^2} y = \frac{1}{x^2}$$

through by $e^{-1/x}$:

$$e^{-1/x} \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{e^{-1/x}}{x^2} y = \frac{e^{-1/x}}{x^2}$$

Now apply the product rule. Since

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(e^{-1/x}\right) = e^{-1/x} \cdot \frac{1}{x^2}$$

we get

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}x}\left(y e^{-1/x}\right) &= e^{-1/x} \frac{\mathrm{d}y}{\mathrm{d}x} + y \frac{\mathrm{d}}{\mathrm{d}x}\left(e^{-1/x}\right) \\ &= e^{-1/x} \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{e^{-1/x}}{x^2} y \end{aligned}$$

The left-hand side is the derivative of $ye^{-1/x}$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(ye^{-1/x}\right) = \frac{e^{-1/x}}{x^2}$$

Integrating both sides with respect to x gives

$$ye^{-1/x} = \int \frac{e^{-1/x}}{x^2}\,\mathrm{d}x + c$$

But

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(e^{-1/x}\right) = \frac{e^{-1/x}}{x^2}$$

so

$$\int \frac{e^{-1/x}}{x^2}\,\mathrm{d}x = e^{-1/x}$$

Hence

$$ye^{-1/x} = e^{-1/x} + c$$

Multiplying through by $e^{1/x}$:

$$y = 1 + ce^{1/x}$$

Now use the condition $y = 3$ when $x = 1$:

$$3 = 1 + ce$$

$$2 = ce$$

$$c = \frac{2}{e}$$

Substitute back:

$$y = 1 + \frac{2}{e}e^{1/x}$$

$$= 1 + 2e^{1/x-1}$$

7. For $x > 0$, solve the differential equation

$$\sinh x \frac{\mathrm{d}y}{\mathrm{d}x} - y \cosh x = \sinh x \cosh x$$

given that $y = 2$ when $x = \ln(1 + \sqrt{2})$.
Give your answer in exact form. **[6]**

Solution

Since $x > 0$, we have $\sinh x > 0$, so we may divide through by $\sinh x$ to put the differential equation into linear form:

$$\sinh x \frac{\mathrm{d}y}{\mathrm{d}x} - y \cosh x = \sinh x \cosh x$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{\cosh x}{\sinh x} y = \cosh x$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \coth x \, y = \cosh x$$

This is a linear differential equation of the form $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$ with $P(x) = -\coth x$.

Using the standard result

$$\int \coth x \, \mathrm{d}x = \ln|\sinh x| + c$$

the integrating factor is

$$\mu(x) = e^{\int P(x) \, \mathrm{d}x}$$

$$= e^{\int -\coth x \, \mathrm{d}x}$$

$$= e^{-\int \coth x \, \mathrm{d}x}$$

$$= e^{-\ln(\sinh x)}$$

Since $x > 0$, we have $\sinh x > 0$, so

$$\mu(x) = \frac{1}{\sinh x}$$

Multiplying the differential equation by this integrating factor gives

$$\frac{1}{\sinh x}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{\coth x}{\sinh x} y = \frac{\cosh x}{\sinh x}$$

The left-hand side is the derivative of $\frac{y}{\sinh x}$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\sinh x}\right) = \coth x$$

Integrating both sides with respect to x:

$$\frac{y}{\sinh x} = \int \coth x \, \mathrm{d}x + c$$

$$= \ln(\sinh x) + c$$

Hence

$$y = \sinh x\big(\ln(\sinh x) + c\big)$$

Now use the condition $y = 2$ when $x = \ln(1 + \sqrt{2})$.

We need $\sinh\big(\ln(1 + \sqrt{2})\big)$. Using

$$\sinh t = \frac{e^t - e^{-t}}{2}$$

with $t = \ln(1 + \sqrt{2})$, we get

$$\sinh\big(\ln(1 + \sqrt{2})\big) = \frac{(1 + \sqrt{2}) - \dfrac{1}{1 + \sqrt{2}}}{2}$$

Now rationalise:

$$\begin{aligned}\frac{1}{1 + \sqrt{2}} &= \frac{1 - \sqrt{2}}{(1 + \sqrt{2})(1 - \sqrt{2})} \\ &= \frac{1 - \sqrt{2}}{1 - 2} \\ &= \sqrt{2} - 1\end{aligned}$$

So

$$\begin{aligned}\sinh\big(\ln(1 + \sqrt{2})\big) &= \frac{(1 + \sqrt{2}) - (\sqrt{2} - 1)}{2} \\ &= \frac{2}{2} \\ &= 1\end{aligned}$$

Substituting into the general solution:

$$\begin{aligned}2 &= 1\big(\ln 1 + c\big) \\ 2 &= 0 + c \\ c &= 2\end{aligned}$$

Therefore the exact solution is

$$y = \sinh x\big(\ln(\sinh x) + 2\big)$$

8. For $x > 0$, solve the differential equation

$$\sinh x \frac{\mathrm{d}y}{\mathrm{d}x} - 2y \cosh x = \sinh^2 x \cosh x$$

given that $y = 5$ when $x = \ln(1 + \sqrt{2})$.
Give your answer in exact form. **[6]**

Solution

Since $x > 0$, we have $\sinh x > 0$, so in particular $\sinh x \neq 0$. We may therefore divide the differential equation by $\sinh x$ to put it into linear form:

$$\begin{aligned}
\sinh x \frac{\mathrm{d}y}{\mathrm{d}x} - 2y \cosh x &= \sinh^2 x \cosh x \\
\frac{\mathrm{d}y}{\mathrm{d}x} - 2\frac{\cosh x}{\sinh x} y &= \sinh x \cosh x \\
\frac{\mathrm{d}y}{\mathrm{d}x} - 2 \coth x \, y &= \sinh x \cosh x
\end{aligned}$$

This is a linear differential equation of the form $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$ with $P(x) = -2 \coth x$.

Using the standard result

$$\int \coth x \, \mathrm{d}x = \ln |\sinh x| + c$$

the integrating factor is

$$\begin{aligned}
\mu(x) &= e^{\int P(x) \, \mathrm{d}x} \\
&= e^{\int -2 \coth x \, \mathrm{d}x} \\
&= e^{-2 \int \coth x \, \mathrm{d}x} \\
&= e^{-2 \ln(\sinh x)}
\end{aligned}$$

Since $x > 0$, we have $\sinh x > 0$, so

$$\mu(x) = \frac{1}{\sinh^2 x}$$

Multiplying the differential equation by this integrating factor gives

$$\frac{1}{\sinh^2 x} \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{2 \coth x}{\sinh^2 x} y = \frac{\sinh x \cosh x}{\sinh^2 x}$$

The left-hand side is the derivative of $\frac{y}{\sinh^2 x}$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\sinh^2 x}\right) = \coth x$$

Now integrate both sides with respect to x:

$$\begin{aligned}
\frac{y}{\sinh^2 x} &= \int \coth x \, \mathrm{d}x \\
&= \ln(\sinh x) + c
\end{aligned}$$

Hence

$$y = \sinh^2 x\big(\ln(\sinh x) + c\big)$$

Now use the condition $y = 5$ when $x = \ln(1 + \sqrt{2})$.

We need $\sinh\big(\ln(1 + \sqrt{2})\big)$. Using

$$\sinh t = \frac{e^t - e^{-t}}{2}$$

with $t = \ln(1 + \sqrt{2})$, we get

$$\begin{aligned}
\sinh\big(\ln(1 + \sqrt{2})\big) &= \frac{(1 + \sqrt{2}) - \dfrac{1}{1 + \sqrt{2}}}{2} \\
&= \frac{(1 + \sqrt{2}) - (\sqrt{2} - 1)}{2} \\
&= \frac{2}{2} \\
&= 1
\end{aligned}$$

So substituting into the solution:

$$\begin{aligned}
5 &= 1^2\big(\ln 1 + c\big) \\
5 &= 0 + c \\
c &= 5
\end{aligned}$$

Therefore the required exact solution is

$$y = \sinh^2 x\big(\ln(\sinh x) + 5\big)$$

9. For $x > 0$, a function y satisfies

$$\sinh x \frac{\mathrm{d}y}{\mathrm{d}x} - y \cosh x = \sinh x \cosh^3 x$$

(a) By first writing the equation in the form $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$, find an integrating factor **[3]**

(b) Hence solve the differential equation, given that $y = 2$ when $x = \ln(1 + \sqrt{2})$ **[6]**

Give your answer in exact form.

Solution

(a) First divide the differential equation by $\sinh x$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{\cosh x}{\sinh x} y = \cosh^3 x$$

This is a linear differential equation of the form $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$ with $P(x) = -\coth x$.
Using the standard result

$$\int \coth x \, \mathrm{d}x = \ln|\sinh x| + c$$

the integrating factor is

$$\begin{aligned} \mu(x) &= e^{\int P(x)\,\mathrm{d}x} \\ &= e^{\int -\coth x \,\mathrm{d}x} \\ &= e^{-\ln(\sinh x)} \end{aligned}$$

Since $x > 0$, we have $\sinh x > 0$, so

$$\mu(x) = \frac{1}{\sinh x}$$

(b) Multiply the equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \coth x \, y = \cosh^3 x$$

by the integrating factor $\frac{1}{\sinh x}$:

$$\frac{1}{\sinh x}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{\cosh x}{\sinh^2 x} y = \frac{\cosh^3 x}{\sinh x}$$

The left-hand side is the derivative of $\frac{y}{\sinh x}$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\sinh x}\right) = \frac{\cosh^3 x}{\sinh x}$$

Integrating both sides gives

$$\frac{y}{\sinh x} = \int \frac{\cosh^3 x}{\sinh x}\,\mathrm{d}x + c$$

Rewrite the integrand using $\cosh^2 x = 1 + \sinh^2 x$:

$$\begin{aligned}\frac{\cosh^3 x}{\sinh x} &= \frac{\cosh x \, \cosh^2 x}{\sinh x} \\ &= \frac{\cosh x(1 + \sinh^2 x)}{\sinh x}\end{aligned}$$

Let

$$u = \sinh x \quad \Longrightarrow \quad \mathrm{d}u = \cosh x \, \mathrm{d}x$$

Then

$$\begin{aligned}\int \frac{\cosh^3 x}{\sinh x} \, \mathrm{d}x &= \int \frac{\cosh x(1 + \sinh^2 x)}{\sinh x} \, \mathrm{d}x \\ &= \int \left(\frac{1}{u} + u\right) \mathrm{d}u \\ &= \ln u + \frac{u^2}{2} \\ &= \ln(\sinh x) + \frac{1}{2}\sinh^2 x\end{aligned}$$

Hence

$$\frac{y}{\sinh x} = \ln(\sinh x) + \frac{1}{2}\sinh^2 x + c$$

and so

$$y = \sinh x \ln(\sinh x) + \frac{1}{2}\sinh^3 x + c \sinh x$$

Use the condition $y = 2$ when $x = \ln(1 + \sqrt{2})$. First find $\sinh\big(\ln(1 + \sqrt{2})\big)$. Using

$$\sinh t = \frac{e^t - e^{-t}}{2}$$

with $t = \ln(1 + \sqrt{2})$, we get

$$\begin{aligned}\sinh\big(\ln(1 + \sqrt{2})\big) &= \frac{e^{\ln(1+\sqrt{2})} - e^{-\ln(1+\sqrt{2})}}{2} \\ &= \frac{(1 + \sqrt{2}) - \dfrac{1}{1 + \sqrt{2}}}{2} \\ &= \frac{(1 + \sqrt{2}) - (\sqrt{2} - 1)}{2} \\ &= 1\end{aligned}$$

So substituting $\sinh x = 1$ and $y = 2$:

$$\begin{aligned}2 &= 1 \times \ln 1 + \frac{1}{2}(1)^3 + c(1) \\ 2 &= 0 + \frac{1}{2} + c \\ c &= \frac{3}{2}\end{aligned}$$

Therefore

$$y = \sinh x \ln(\sinh x) + \frac{1}{2}\sinh^3 x + \frac{3}{2}\sinh x$$

10. **(a)** Show that, for $0 < x < \frac{\pi}{2}$, an appropriate integrating factor for

$$\cos x \frac{\mathrm{d}y}{\mathrm{d}x} + y \sin x = x$$

is $\sec x$ **[3]**

(b) Hence find the solution of

$$\cos x \frac{\mathrm{d}y}{\mathrm{d}x} + y \sin x = x$$

for $0 < x < \frac{\pi}{2}$, given that $y = 1$ when $x = \frac{\pi}{4}$ **[6]**

Solution

(a) First divide the differential equation by $\cos x$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + y \tan x = x \sec x$$

This is a linear differential equation of the form $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$ with $P(x) = \tan x$.
Using the standard result

$$\int \tan x \, \mathrm{d}x = \ln |\sec x| + c$$

the integrating factor is

$$\begin{aligned} \mu(x) &= e^{\int P(x) \, \mathrm{d}x} \\ &= e^{\int \tan x \, \mathrm{d}x} \\ &= e^{\ln(\sec x)} \end{aligned}$$

Since $0 < x < \frac{\pi}{2}$, we have $\sec x > 0$, so

$$\mu(x) = \sec x$$

Therefore an appropriate integrating factor is $\sec x$.

(b) Using the result from part (a), the equation is

$$\frac{\mathrm{d}y}{\mathrm{d}x} + y \tan x = x \sec x$$

Multiply through by the integrating factor $\sec x$:

$$\sec x \frac{\mathrm{d}y}{\mathrm{d}x} + y \sec x \tan x = x \sec^2 x$$

The left-hand side is the derivative of $y \sec x$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}(y \sec x) = x \sec^2 x$$

Integrating both sides gives

$$y \sec x = \int x \sec^2 x \, \mathrm{d}x + c$$

Integrate by parts with

$$u = x, \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \sec^2 x$$

so that

$$\frac{\mathrm{d}u}{\mathrm{d}x} = 1, \qquad v = \tan x$$

Using integration by parts:

$$\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$

Then

$$\begin{aligned}\int x\sec^2 x\,\mathrm{d}x &= x\tan x - \int \tan x\,\mathrm{d}x \\ &= x\tan x - (-\ln(\cos x)) \\ &= x\tan x + \ln(\cos x)\end{aligned}$$

Therefore

$$y\sec x = x\tan x + \ln(\cos x) + c$$

Now multiply through by $\cos x$:

$$\begin{aligned}y &= \cos x(x\tan x + \ln(\cos x) + c) \\ &= x\sin x + \cos x\ln(\cos x) + c\cos x\end{aligned}$$

Use the condition $y = 1$ when $x = \frac{\pi}{4}$. Substituting into

$$y\sec x = x\tan x + \ln(\cos x) + c$$

gives

$$\begin{aligned}1 \times \sec\frac{\pi}{4} &= \frac{\pi}{4}\tan\frac{\pi}{4} + \ln\left(\cos\frac{\pi}{4}\right) + c \\ \sqrt{2} &= \frac{\pi}{4} + \ln\left(\frac{\sqrt{2}}{2}\right) + c \\ \sqrt{2} &= \frac{\pi}{4} - \frac{1}{2}\ln 2 + c\end{aligned}$$

So

$$c = \sqrt{2} - \frac{\pi}{4} + \frac{1}{2}\ln 2$$

Substitute this into the expression for y:

$$y = x\sin x + \cos x\left[\ln(\cos x) + \sqrt{2} - \frac{\pi}{4} + \frac{1}{2}\ln 2\right]$$

11. Find the general solution of the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \cot x\, y = \frac{\cos x}{\sin^2 x \left(1 + [\ln(\sin x)]^2\right)}$$

where $0 < x < \pi$ **[5]**

Solution

This is a linear differential equation of the form $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$ with $P(x) = \cot x$.

Using the standard result

$$\int \cot x \,\mathrm{d}x = \ln|\sin x| + c$$

the integrating factor is

$$\begin{aligned}\mu(x) &= e^{\int P(x)\,\mathrm{d}x} \\ &= e^{\int \cot x\,\mathrm{d}x} \\ &= e^{\ln(\sin x)}\end{aligned}$$

Since $0 < x < \pi$, we have $\sin x > 0$, so

$$\mu(x) = \sin x$$

Multiply the whole differential equation by $\sin x$:

$$\sin x\frac{\mathrm{d}y}{\mathrm{d}x} + \sin x \cot x\, y = \sin x \cdot \frac{\cos x}{\sin^2 x\left(1 + [\ln(\sin x)]^2\right)}$$

$$\sin x\frac{\mathrm{d}y}{\mathrm{d}x} + \cos x\, y = \frac{\cos x}{\sin x\left(1 + [\ln(\sin x)]^2\right)}$$

The left-hand side is the derivative of $y \sin x$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}(y\sin x) = \frac{\cos x}{\sin x\left(1 + [\ln(\sin x)]^2\right)}$$

Integrating both sides gives

$$\int \frac{\mathrm{d}}{\mathrm{d}x}(y\sin x)\,\mathrm{d}x = \int \frac{\cos x}{\sin x\left(1 + [\ln(\sin x)]^2\right)}\,\mathrm{d}x$$

For the right-hand side, let

$$u = \ln(\sin x)$$

Then

$$\begin{aligned}\frac{\mathrm{d}u}{\mathrm{d}x} &= \frac{1}{\sin x} \cdot \cos x \\ &= \frac{\cos x}{\sin x}\end{aligned}$$

Hence

$$\mathrm{d}u = \frac{\cos x}{\sin x}\,\mathrm{d}x$$

So

$$\begin{aligned}\int \frac{\cos x}{\sin x\Big(1+[\ln(\sin x)]^2\Big)}\,\mathrm{d}x &= \int \frac{1}{1+u^2}\,\mathrm{d}u \\ &= \arctan(u) + c \\ &= \arctan(\ln(\sin x)) + c\end{aligned}$$

Therefore

$$y\sin x = \arctan(\ln(\sin x)) + c$$

Finally, divide by $\sin x$:

$$y = \frac{\arctan(\ln(\sin x)) + c}{\sin x}$$

12. **(a)** Show that an appropriate integrating factor for

$$x(x-1)\frac{\mathrm{d}y}{\mathrm{d}x}+(x+1)y=x(x-1)^2$$

is $\frac{(x-1)^2}{x}$ **[5]**

(b) Hence find the solution of the differential equation

$$x(x-1)\frac{\mathrm{d}y}{\mathrm{d}x}+(x+1)y=x(x-1)^2$$

for $x>1$, given that $y=2$ when $x=2$ **[6]**

Solution

(a) First write the differential equation in linear form by dividing through by $x(x-1)$:

$$\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{x+1}{x(x-1)}y=x-1$$

This is a linear differential equation of the form $\frac{\mathrm{d}y}{\mathrm{d}x}+P(x)y=Q(x)$ with $P(x)=\frac{x+1}{x(x-1)}$.

Decompose into partial fractions:

$$\frac{x+1}{x(x-1)}=\frac{A}{x}+\frac{B}{x-1}$$

$$x+1=A(x-1)+Bx$$

$$x+1=(A+B)x-A$$

Equating coefficients gives

$$A+B=1, \qquad -A=1$$

so

$$A=-1, \qquad B=2$$

Hence

$$\frac{x+1}{x(x-1)}=-\frac{1}{x}+\frac{2}{x-1}$$

The integrating factor is

$$\mu(x)=e^{\int P(x)\,\mathrm{d}x}$$

$$=e^{\int\left(-\frac{1}{x}+\frac{2}{x-1}\right)\mathrm{d}x}$$

$$=e^{-\ln x+2\ln(x-1)}$$

Since $x>1$, we have $x>0$ and $x-1>0$, so an appropriate integrating factor is

$$\mu(x)=\frac{(x-1)^2}{x}$$

(b) Using the integrating factor from part (a):

$$\mu(x) = \frac{(x-1)^2}{x}$$

Multiply the linear form

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{x+1}{x(x-1)}y = x - 1$$

through by $\mu(x)$:

$$\frac{(x-1)^2}{x}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{(x-1)^2}{x} \cdot \frac{x+1}{x(x-1)}\, y = \frac{(x-1)^2}{x}(x-1)$$

The left-hand side is the derivative of $\frac{(x-1)^2}{x}y$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{(x-1)^2}{x}y\right) = \frac{(x-1)^3}{x}$$

Integrating both sides:

$$\frac{(x-1)^2}{x}y = \int \frac{(x-1)^3}{x}\,\mathrm{d}x$$

Expand the integrand:

$$\begin{aligned}\frac{(x-1)^3}{x} &= \frac{x^3 - 3x^2 + 3x - 1}{x} \\ &= x^2 - 3x + 3 - \frac{1}{x}\end{aligned}$$

Hence

$$\begin{aligned}\frac{(x-1)^2}{x}y &= \int \left(x^2 - 3x + 3 - \frac{1}{x}\right)\mathrm{d}x \\ &= \frac{x^3}{3} - \frac{3x^2}{2} + 3x - \ln x + c\end{aligned}$$

So

$$y = \frac{x}{(x-1)^2}\left(\frac{x^3}{3} - \frac{3x^2}{2} + 3x - \ln x + c\right)$$

Now use the condition $y = 2$ when $x = 2$:

$$\begin{aligned}2 &= \frac{2}{(2-1)^2}\left(\frac{2^3}{3} - \frac{3(2^2)}{2} + 3(2) - \ln 2 + c\right) \\ 2 &= 2\left(\frac{8}{3} - 6 + 6 - \ln 2 + c\right) \\ 2 &= 2\left(\frac{8}{3} - \ln 2 + c\right)\end{aligned}$$

Divide by 2:

$$1 = \frac{8}{3} - \ln 2 + c$$

So

$$c = 1 - \frac{8}{3} + \ln 2 = \ln 2 - \frac{5}{3}$$

Substituting back:

$$y = \frac{x}{(x-1)^2}\left(\frac{x^3}{3} - \frac{3x^2}{2} + 3x - \ln x + \ln 2 - \frac{5}{3}\right)$$

13. **(a)** Show that

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(\ln(\cosh x+1)\big)=\tanh\Big(\frac{x}{2}\Big) \qquad \textbf{[3]}$$

(b) Hence solve the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x}+\tanh\Big(\frac{x}{2}\Big)\,y=\sinh x$$

for which $y=2$ when $x=0$. Give your answer in exact form **[5]**

Solution

(a) Using $\frac{\mathrm{d}}{\mathrm{d}x}\ln u=\frac{1}{u}\frac{\mathrm{d}u}{\mathrm{d}x}$:

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(\ln(\cosh x+1)\big)=\frac{1}{\cosh x+1}\cdot\frac{\mathrm{d}}{\mathrm{d}x}(\cosh x+1)=\frac{\sinh x}{\cosh x+1}$$

Now use the double angle identities with x replaced by $\frac{x}{2}$:

$$\sinh x=2\sinh\Big(\frac{x}{2}\Big)\cosh\Big(\frac{x}{2}\Big),\qquad \cosh x+1=2\cosh^2\Big(\frac{x}{2}\Big)$$

So

$$\begin{aligned}\frac{\sinh x}{\cosh x+1}&=\frac{2\sinh\left(\frac{x}{2}\right)\cosh\left(\frac{x}{2}\right)}{2\cosh^2\left(\frac{x}{2}\right)}\\&=\frac{\sinh\left(\frac{x}{2}\right)}{\cosh\left(\frac{x}{2}\right)}\\&=\tanh\Big(\frac{x}{2}\Big)\end{aligned}$$

Hence

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(\ln(\cosh x+1)\big)=\tanh\Big(\frac{x}{2}\Big)$$

as required.

(b) This is a linear differential equation of the form $\frac{\mathrm{d}y}{\mathrm{d}x}+P(x)y=Q(x)$ with $P(x)=\tanh\left(\frac{x}{2}\right)$.
From part (a):

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(\ln(\cosh x+1)\big)=\tanh\Big(\frac{x}{2}\Big)$$

The integrating factor is

$$\begin{aligned}\mu(x)&=e^{\int P(x)\,\mathrm{d}x}\\&=e^{\int\tanh\left(\frac{x}{2}\right)\mathrm{d}x}\\&=e^{\ln(\cosh x+1)}\\&=\cosh x+1\end{aligned}$$

Multiplying the differential equation throughout by $\cosh x+1$:

$$(\cosh x+1)\frac{\mathrm{d}y}{\mathrm{d}x}+(\cosh x+1)\tanh\Big(\frac{x}{2}\Big)y=(\cosh x+1)\sinh x$$

The left-hand side is the derivative of $(\cosh x + 1)y$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\big((\cosh x + 1)y\big) = (\cosh x + 1)\sinh x$$

We can integrate the right-hand side by reversing the chain rule:

$$\int (\cosh x + 1)\sinh x \,\mathrm{d}x = \frac{1}{2}(\cosh x + 1)^2 + c$$

So the differential equation becomes

$$\begin{aligned}(\cosh x + 1)y &= \int (\cosh x + 1)\sinh x \,\mathrm{d}x \\ &= \frac{1}{2}(\cosh x + 1)^2 + c\end{aligned}$$

Therefore

$$y = \frac{1}{2}(\cosh x + 1) + \frac{c}{\cosh x + 1}$$

Use the condition $y = 2$ when $x = 0$. Since $\cosh 0 = 1$, this gives

$$\begin{aligned}2 &= \frac{1}{2}(1 + 1) + \frac{c}{1 + 1} \\ 2 &= 1 + \frac{c}{2}\end{aligned}$$

Hence

$$c = 2$$

Substituting back:

$$y = \frac{\cosh x + 1}{2} + \frac{2}{\cosh x + 1}$$

Alternative Solution:

The integral can also be found with the double angle formula for $\sinh 2x$:

$$\int (\cosh x + 1)\sinh x \,\mathrm{d}x = \int (\cosh x \sinh x + \sinh x)\,\mathrm{d}x = \int \left(\frac{1}{2}\sinh 2x + \sinh x\right)\mathrm{d}x$$

This leads to an equivalent form of the final answer:

$$y = \frac{\frac{1}{4}\cosh 2x + \cosh x + \frac{11}{4}}{\cosh x + 1}$$

14. **(a)** Show that, for $x > 0$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\coth\frac{x}{2}\right)\right) = -\operatorname{cosech} x \qquad \textbf{[3]}$$

(b) Hence solve the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \operatorname{cosech} x\, y = \tanh\frac{x}{2}, \qquad x > 0$$

for which $y = 1$ when $x = \ln 3$. Give your answer in exact form **[5]**

Solution

(a) Let

$$u = \frac{x}{2}$$

Then, by the chain rule:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\coth\frac{x}{2}\right)\right) &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\ln(\coth u)\right)\\ &= \frac{1}{\coth u}\cdot\frac{\mathrm{d}}{\mathrm{d}x}(\coth u)\\ &= \frac{1}{\coth u}\cdot\left(-\operatorname{cosech}^2 u\right)\cdot\frac{1}{2}\end{aligned}$$

Now simplify:

$$\begin{aligned}\frac{1}{\coth u}\cdot\left(-\operatorname{cosech}^2 u\right)\cdot\frac{1}{2} &= -\frac{1}{2}\cdot\frac{\operatorname{cosech}^2 u}{\coth u}\\ &= -\frac{1}{2}\cdot\frac{1/\sinh^2 u}{\cosh u/\sinh u}\\ &= -\frac{1}{2\sinh u\cosh u}\end{aligned}$$

Using $\sinh(2u) = 2\sinh u\cosh u$:

$$\begin{aligned}-\frac{1}{2\sinh u\cosh u} &= -\frac{1}{\sinh(2u)}\\ &= -\frac{1}{\sinh x}\\ &= -\operatorname{cosech} x\end{aligned}$$

Therefore

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\coth\frac{x}{2}\right)\right) = -\operatorname{cosech} x$$

as required.

(b) This is a linear differential equation of the form $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$ with $P(x) = -\operatorname{cosech} x$.

The integrating factor is

$$\begin{aligned}\mu(x) &= e^{\int P(x)\,\mathrm{d}x} \\ &= e^{\int -\operatorname{cosech} x\,\mathrm{d}x} \\ &= e^{\ln\left(\coth\frac{x}{2}\right)} \\ &= \coth\frac{x}{2}\end{aligned}$$

using part (a).

Next, from part (a):

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\coth\frac{x}{2}\right)\right) = -\operatorname{cosech} x$$

so

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left(\coth\frac{x}{2}\right) &= \coth\frac{x}{2}\cdot\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\coth\frac{x}{2}\right)\right) \\ &= -\operatorname{cosech} x\,\coth\frac{x}{2}\end{aligned}$$

Multiply the differential equation by $\coth\frac{x}{2}$:

$$\coth\frac{x}{2}\,\frac{\mathrm{d}y}{\mathrm{d}x} - \operatorname{cosech} x\,\coth\frac{x}{2}\,y = \coth\frac{x}{2}\tanh\frac{x}{2}$$

The left-hand side is the derivative of $y\coth\frac{x}{2}$ and the right-hand side is 1, since $\coth\frac{x}{2}\tanh\frac{x}{2} = 1$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(y\coth\frac{x}{2}\right) = 1$$

Integrating:

$$y\coth\frac{x}{2} = \int 1\,\mathrm{d}x = x + c$$

So

$$y = (x+c)\tanh\frac{x}{2}$$

Now use $y = 1$ when $x = \ln 3$:

$$1 = (\ln 3 + c)\tanh\frac{\ln 3}{2}$$

Evaluate $\tanh\frac{\ln 3}{2}$. Using

$$\tanh t = \frac{e^t - e^{-t}}{e^t + e^{-t}}$$

with $t = \frac{\ln 3}{2}$, we get

$$\begin{aligned}\tanh \frac{\ln 3}{2} &= \frac{e^{\frac{1}{2}\ln 3} - e^{-\frac{1}{2}\ln 3}}{e^{\frac{1}{2}\ln 3} + e^{-\frac{1}{2}\ln 3}} \\ &= \frac{e^{\ln 3} - 1}{e^{\ln 3} + 1} \\ &= \frac{3-1}{3+1} \\ &= \frac{1}{2}\end{aligned}$$

Therefore

$$\begin{aligned}1 &= (\ln 3 + c) \times \frac{1}{2} \\ 2 &= \ln 3 + c \\ c &= 2 - \ln 3\end{aligned}$$

Substituting back gives

$$y = (x + 2 - \ln 3) \tanh \frac{x}{2}$$

15. **(a)** Show that

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\frac{\cosh x+1}{\sinh x}\right)\right)=-\operatorname{cosech} x \qquad \textbf{[3]}$$

(b) Hence solve, for $x>0$,

$$\sinh x\,\frac{\mathrm{d}y}{\mathrm{d}x}-y=1-\cosh x$$

given that $y=2$ when $x=\ln 3$. Give your answer in exact form **[5]**

Solution

(a) Write

$$\ln\left(\frac{\cosh x+1}{\sinh x}\right)=\ln(\cosh x+1)-\ln(\sinh x)$$

Differentiate term by term:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\frac{\cosh x+1}{\sinh x}\right)\right)&=\frac{\sinh x}{\cosh x+1}-\frac{\cosh x}{\sinh x}\\&=\frac{\sinh^2 x-\cosh x(\cosh x+1)}{\sinh x(\cosh x+1)}\end{aligned}$$

Now simplify the numerator using $\cosh^2 x-\sinh^2 x=1$:

$$\begin{aligned}\sinh^2 x-\cosh x(\cosh x+1)&=\sinh^2 x-\cosh^2 x-\cosh x\\&=-(\cosh^2 x-\sinh^2 x)-\cosh x\\&=-1-\cosh x\\&=-(\cosh x+1)\end{aligned}$$

So

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\frac{\cosh x+1}{\sinh x}\right)\right)&=\frac{-(\cosh x+1)}{\sinh x(\cosh x+1)}\\&=-\frac{1}{\sinh x}\\&=-\operatorname{cosech} x\end{aligned}$$

Hence

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\frac{\cosh x+1}{\sinh x}\right)\right)=-\operatorname{cosech} x$$

(b) Start with

$$\sinh x\,\frac{\mathrm{d}y}{\mathrm{d}x}-y=1-\cosh x$$

Since $x>0$, we have $\sinh x\neq 0$, so divide through by $\sinh x$:

$$\frac{\mathrm{d}y}{\mathrm{d}x}-\operatorname{cosech} x\,y=\frac{1-\cosh x}{\sinh x}$$

This is a linear differential equation of the form $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$ with $P(x) = -\operatorname{cosech} x$.

From part (a):

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\frac{\cosh x + 1}{\sinh x}\right)\right) = -\operatorname{cosech} x$$

The integrating factor is

$$\begin{aligned}\mu(x) &= e^{\int P(x)\,\mathrm{d}x} \\ &= e^{\int -\operatorname{cosech} x\,\mathrm{d}x} \\ &= e^{\ln\left(\frac{\cosh x+1}{\sinh x}\right)} \\ &= \frac{\cosh x + 1}{\sinh x}\end{aligned}$$

Multiply the differential equation throughout by $\mu(x)$:

$$\frac{\cosh x + 1}{\sinh x}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{\cosh x + 1}{\sinh^2 x}y = \frac{(\cosh x + 1)(1 - \cosh x)}{\sinh^2 x}$$

Simplify the right-hand side:

$$\begin{aligned}\frac{(\cosh x + 1)(1 - \cosh x)}{\sinh^2 x} &= \frac{1 - \cosh^2 x}{\sinh^2 x} \\ &= \frac{-\sinh^2 x}{\sinh^2 x} \\ &= -1\end{aligned}$$

The left-hand side is the derivative of $\frac{\cosh x+1}{\sinh x}y$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\cosh x + 1}{\sinh x}y\right) = -1$$

Integrate with respect to x:

$$\frac{\cosh x + 1}{\sinh x}y = -x + c$$

Hence

$$y = \frac{(c - x)\sinh x}{\cosh x + 1}$$

Now use the condition $y = 2$ when $x = \ln 3$.

First find $\sinh(\ln 3)$ and $\cosh(\ln 3)$. Using

$$\sinh t = \frac{e^t - e^{-t}}{2}, \qquad \cosh t = \frac{e^t + e^{-t}}{2}$$

with $t = \ln 3$, we get

$$\sinh(\ln 3) = \frac{e^{\ln 3} - e^{-\ln 3}}{2} = \frac{3 - \frac{1}{3}}{2} = \frac{4}{3}$$

$$\cosh(\ln 3) = \frac{e^{\ln 3} + e^{-\ln 3}}{2} = \frac{3 + \frac{1}{3}}{2} = \frac{5}{3}$$

Substitute into the solution:

$$\begin{aligned} 2 &= \frac{(c - \ln 3)\sinh(\ln 3)}{\cosh(\ln 3) + 1} \\ &= \frac{(c - \ln 3) \times \frac{4}{3}}{\frac{5}{3} + 1} \\ &= \frac{(c - \ln 3) \times \frac{4}{3}}{\frac{8}{3}} \\ &= \frac{c - \ln 3}{2} \end{aligned}$$

Therefore

$$\begin{aligned} 2 &= \frac{c - \ln 3}{2} \\ 4 &= c - \ln 3 \\ c &= 4 + \ln 3 \end{aligned}$$

Substituting back:

$$y = \frac{(4 + \ln 3 - x)\sinh x}{\cosh x + 1}$$